

NY-ALESUND, Svalbard and Jan Mayen

WMO: 010070

Lat: 78.9228N

Lon: 11.9331E

Elev: 25

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-13.3	-8.8	-21.6	1.7	-11.5	-18.2	2.0	-7.4	40.5	19.7	35.7	20.4	5.2	130	0.937

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.3	49.8	44.3	47.7	43.0	46.2	42.0	45.1	48.7	43.7	47.1	42.4	45.7	9.0	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
42.4	39.9	46.2	40.8	37.5	44.9	39.5	35.6	43.9	17.7	48.9	17.0	47.3	16.4	45.9	55.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 32.4	27.8	23.7	DB	-12.6	54.3	6.2	3.8	-17.0	57.1	-20.7	59.4	-24.2	61.5	-28.7	64.3	(4)
(5)			WB	-13.3	48.0	5.9	3.1	-17.6	50.3	-21.0	52.1	-24.3	53.9	-28.6	56.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.2	15.5	14.2	11.0	16.7	28.4	37.3	42.7	40.3	34.3	25.6	19.6	16.1
	DBStd	13.59	12.18	11.32	10.54	10.18	5.92	3.85	3.17	3.66	5.81	8.14	8.92	10.09
	HDD50	9051	1069	1001	1208	1000	671	382	228	302	472	756	912	1050
	HDD65	14525	1534	1421	1673	1450	1136	832	692	767	922	1221	1362	1515
	CDD50	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	8.8	10.7	10.7	10.0	8.6	7.9	6.9	6.6	6.5	7.7	9.0	10.2	11.2
Precipitation	PrecAvg	18.6	2.4	1.6	1.8	1.1	0.7	0.6	1.1	1.5	2.3	1.6	2.1	1.8
	PrecMax	29.5	8.5	4.9	5.2	4.2	2.7	2.2	3.4	3.9	5.4	5.4	11.3	4.5
	PrecMin	10.6	0.2	0.1	0.4	0.4	0.0	0.1	0.1	0.2	0.3	0.0	0.2	0.2
	PrecStd	4.6	2.0	1.3	1.2	0.8	0.5	0.5	0.8	0.9	1.5	1.3	2.1	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.9	38.3	36.6	38.9	41.9	48.0	55.1	52.6	48.3	44.1	39.6	41.0
		MCWB	35.5	34.5	33.0	35.3	36.0	42.4	48.8	45.4	44.9	40.5	35.9	37.2
	2%	DB	35.6	35.5	33.3	35.9	38.7	45.8	51.1	48.7	45.7	41.3	36.9	35.5
		MCWB	32.8	32.7	30.6	33.0	34.7	41.0	45.3	43.8	42.1	38.1	34.1	32.1
	5%	DB	33.5	32.5	30.4	33.3	37.0	44.1	48.7	46.8	43.9	39.2	34.8	33.0
		MCWB	30.9	30.1	28.1	30.7	33.4	39.8	43.6	42.5	40.5	36.2	32.2	30.4
	10%	DB	31.0	29.8	26.6	31.0	35.6	42.5	47.0	45.3	42.2	37.1	32.2	30.5
		MCWB	28.7	27.6	24.6	28.6	32.3	38.5	42.5	41.3	39.3	34.6	29.8	28.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.8	35.0	33.6	35.8	36.8	43.4	49.6	46.2	45.2	41.1	36.9	37.5
		MCDB	38.3	37.8	36.1	38.4	39.8	46.9	54.0	50.2	47.6	43.8	39.1	41.2
	2%	WB	33.1	32.9	30.7	33.3	35.3	41.5	46.0	44.4	42.6	38.6	34.3	32.6
		MCDB	35.3	35.3	33.3	35.5	38.4	44.9	49.7	48.3	45.1	41.1	36.6	34.8
	5%	WB	31.0	30.3	28.0	31.0	33.8	40.3	44.2	43.0	40.8	36.5	32.3	30.7
		MCDB	33.3	32.4	30.2	33.2	36.6	43.5	47.7	46.4	43.3	38.9	34.5	32.8
	10%	WB	28.7	27.6	24.8	28.7	32.4	39.0	43.0	41.7	39.4	34.6	30.0	28.3
		MCDB	31.0	29.8	26.5	30.9	35.2	41.9	46.4	44.8	42.1	37.0	32.1	30.8
Mean Daily Temperature Range	5% DB	MDBR	8.9	9.5	9.5	9.5	6.9	5.1	5.3	5.3	5.9	7.2	8.3	9.1
		MCDBR	9.5	10.5	11.3	8.6	7.2	6.8	7.6	7.1	7.8	7.7	9.5	10.4
		MCWBR	8.4	9.5	10.3	7.4	5.0	4.6	5.0	4.7	6.3	6.6	8.4	9.5
	5% WB	MCDBR	9.4	11.2	11.3	8.9	6.8	6.3	7.2	6.8	7.6	7.8	10.3	10.8
		MCWBR	8.5	10.2	10.4	7.8	5.2	4.5	5.2	4.7	6.3	6.8	9.3	10.2
Clear-Sky Solar Irradiance	taub	N/A	N/A	0.250	0.297	0.315	0.306	0.308	0.294	0.243	N/A	N/A	N/A	N/A
	taud	N/A	N/A	2.038	2.132	2.174	2.293	2.363	2.367	2.272	N/A	N/A	N/A	N/A
	Ebn at Noon	0	0	166	231	252	264	258	238	186	0	0	0	0
	Edh at Noon	0	0	16	27	32	30	26	22	12	0	0	0	0
All-Sky Solar Radiation	RadAvg	0	0	202	806	1423	1737	1428	900	315	0	0	0	0
	RadStd	0	0	17	82	119	139	118	73	32	0	0	0	0

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47) Regional (3 neighbors)	+2.74	+8.33	+7.38	N/A	N/A	N/A	-994	-988	N/A	N/A	(47)

Nomenclature: See separate page